Natural Language Processing.

→ making computers understand natural language.

→ there is no standard mathematical definition of text.

→ Language → mathematical

Use cases: document categorization.

Question → Answer. ✓

Translate ✓

Summarize ✓

Generate text ✓

Named Entity.

Text → codes

Python Code → Matlab

image → text. (IC).

text → image (SD).

→ 1) Text → vectors (Logical). / Context → Embedding

→ 2) vectors → models → o/p. (what models). → Prediction / Sequence models.

→ words -> vectors (frequency based embeddings).
→
→ words -> vectors (contextualized embedding).

→ Recurrent neural network
→ Long Short term network =
→ GRU
→ Transformers, BERT.
→ NER
→ Translation / Q-A

Definitions:

Document: a document, a 'row' of text data.
Corpus: Collection of documents that you are going to consider for your task.

Token: Smallest unit for which you want to derive the context → (word). in a document.
| sentence
--- → Token.

word

Running. word
Run ning.

Sub word embedding.

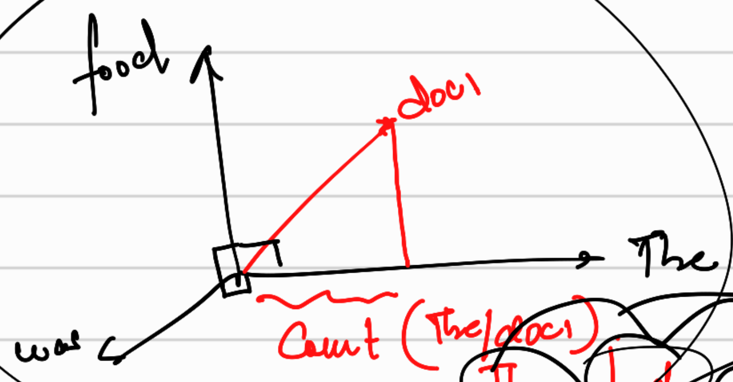
Vocabulary: Union of tokens

Hugging Face

Frequency embeddings:

Sentiment analysis.

		Target
doc1:	The food was amazing	+ve.
doc2:	The ambience was poor.	-ve.



10,000

	The	food	was	amazing	ambience	poor	
→ doc1 (Tweet)	1	1	1	1	0	0	+ve
→ doc2	1	0	1	0	1	1	-ve

~~doc~~ 1,01,000

Document Term matrix

Bag of word. $y = f(x)$.

Count vectorization.

Problems: → Sparse matrix → PCA.

↳ Remove unnecessary features/tokens.

Regular expressions Regex %, $, !, numbers, spc...

Stemming & lemmatization Caring → Care

⊙. Case Reconciliation.

⊙. Remove stop words

⊙ No respect towards sequence.

θ

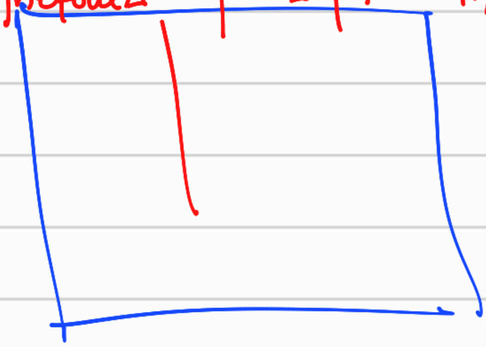
The food is pathetic ambience was great.

The food was great, ambience was Pathetic

Solve

Ngrams

The food is food is path is p---



Context missing

The food was great.

I had an amazing dinner.